Further, it may be noted that the odds are $a : b$ in favour of an event is the same as to say that the odds are $b : a$ against the event.

If the probability of an event is p, then the odds in favour of its occurrence are p to $(1-p)$ and the odds against its occurrence are $(1-p)$ to p.

Illustration 12.7

(i) Suppose a die is rolled.

Let S be the sample space and A be the event of getting 5.

$n(S) = 6$, $n(A) = 1$ and $n(\bar{A}) = 5$.

It can also be interpreted as

Odds in favour of A is $1:5$ or $\frac{1}{5}$, odds against A is $5:1$ or $\frac{5}{1}$,

and $P(A) = \frac{n(A)}{n(A)+n(\bar{A})} = \frac{1}{5+1} = \frac{1}{6} = \frac{n(A)}{n(S)}$.

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(ii) Suppose B is an event such that odds in **favour** of B is 3:5, then $P(B) = \frac{3}{8}$

(iii) Suppose C is an event such that odds **against** C is 4:11, then $P(C) = \frac{11}{15}$.

Example 12.11

A man has 2 ten rupee notes, 4 hundred rupee notes and 6 five hundred rupee notes in his pocket. If 2 notes are taken at random, what are the odds in favour of both notes being of hundred rupee denomination and also its probability?

Solution

Let S be the sample space and A be the event of taking 2 hundred rupee note.

Therefore, $n(S) = 12c_2 = 66$, $n(A) = 4c_2 = 6$ and $n(\bar{A}) = 66 - 6 = 60$

Therefore, odds in favour of A is 6: 60

That is, odds in favour of A is 1: 10, and $P(A) = \frac{1}{11}$.

EXERCISE 12.1

(1) An experiment has the four possible mutually exclusive and exhaustive outcomes A, B, C, and D. Check whether the following assignments of probability are permissible.

(i) $P(A) = 0.15$, $P(B) = 0.30$, $P(C) = 0.43$, $P(D) = 0.12$

(ii) $P(A) = 0.22$, $P(B) = 0.38$, $P(C) = 0.16$, $P(D) = 0.34$

(iii) $P(A) = \frac{2}{5}$, $P(B) = \frac{3}{5}$, $P(C) = -\frac{1}{5}$, $P(D) = \frac{1}{5}$

(2) If two coins are tossed simultaneously, then find the probability of getting

(i) one head and one tail (ii) atmost two tails

(3) Five mangoes and 4 apples are in a box. If two fruits are chosen at random, find the probability that (i) one is a mango and the other is an apple (ii) both are of the same variety.

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